Auditing AI-Based ESG Scores: Transparency, Bias, Robustness, and Sustainability Claims

1. INTRODUCTION

AI-based ESG scoring tools are increasingly used by investors and regulators. However, concerns remain about transparency, fairness, robustness, and potential bias. This study evaluates AI-generated ESG risk scores using two public datasets and multiple machine learning models.

2.Datasets

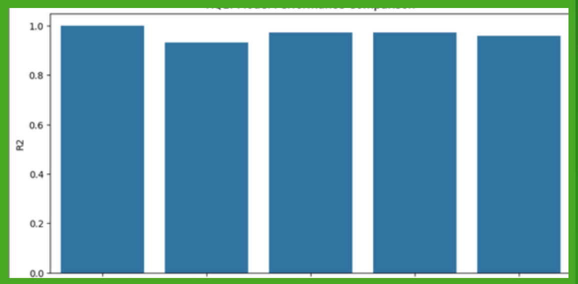
Dataet 1: Public Company ESG Risk Ratings(Kaggle)

1. -7000 companies
2. -ESG risk scores: Environment , social, Governance , Controversy

Dataset 2 : S&P 500 ESG Sustainability Reports(kaggle)

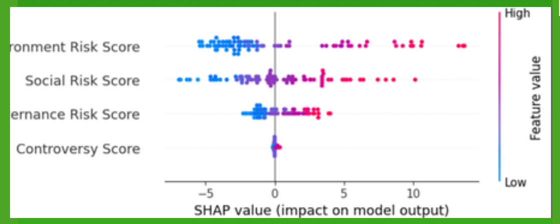
1. -5,000 reports
2. text + ESG scores

RQ1:which ML model predicts the scores most accurately?



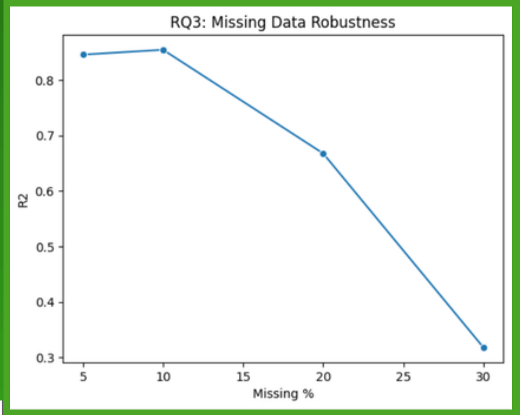
Linear Regression achieved the highest R^2 score, demonstrarting superior predictive performance for ESG risk Scores.

RQ2:Which ESG Factors Most Influence AI-generated ESG scores?



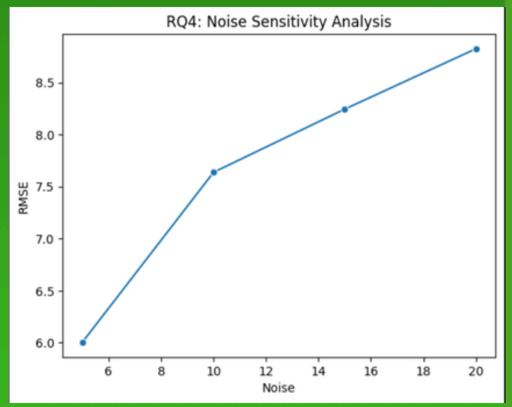
Social risk has the greatest influence on ESG risk Prediction, Ffolowed by Governanace

RQ3:How robust are ESG predictions when sustainability data is missing?



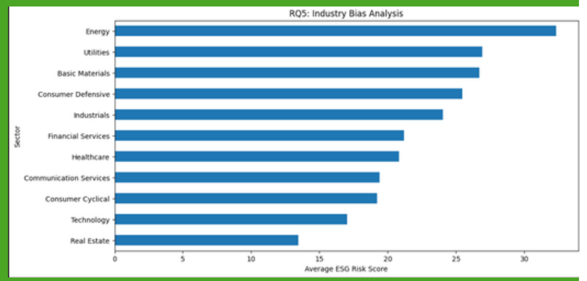
Model performance gradually declines as missing data increases, but remains reasonably robust up to 20% missing data.

RQ4:How Sensitive Are ESG Predictions to Noisy Sustainability Data?



Prediction error (RMSE) increases consistently with higher noise levels, indicating sensitivity to noisy sustainability data.

RQ5:Does the AI Model Exhibit Industry-Level Bias?



Significant variation exists across sectors. Energy shows the highest ESG risk, indicating potential industry-level bias.

3.Methodology

Data Collection → Pre-Processing → Model training(ml-models) → Evaluation-audit using (RQ1-RQ5) → Insights and Recommendation

CONCLUSION

Our audit framework provides a holistic view of AI-based ESG scoring systems. While models are effective, they are sensitive to data quality and show industry-level variations. Integrating explainability, robustness checks, and fairness audits is essential for building trustworthy, transparent, and accountable ESG intelligence systems.